# Practice workbook · Compare DNA with RNA

Contrasting strands, sugar and bases of DNA and RNA, naming RNA types and locations, and using structure or base data to identify a nucleic acid.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-COMPARE-DNA-RNA

# 1. Question 4 · part a.iii · 3 marks

November 2023 Paper 1

Worked-example retry · Attempt part a.iii only.

**4** The mass of DNA in a body cell changes during the cell cycle.

(a) (i) Which structure in a cell contains DNA?

**(1)**

- ☒ **A** chromosome
- ☒ **B** endoplasmic reticulum
- ☒ **C** cell membrane
- ☒ **D** ribosome

(ii) Name **one** other type of nucleic acid.

**(1)**

(iii) Give **three** differences between DNA and the nucleic acid that you have named in (a)(ii).

**(3)**

| Differences | DNA | Named nucleic acid |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |

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# 1. Question 4 · part a.iii · 3 marks

November 2023 Paper 1

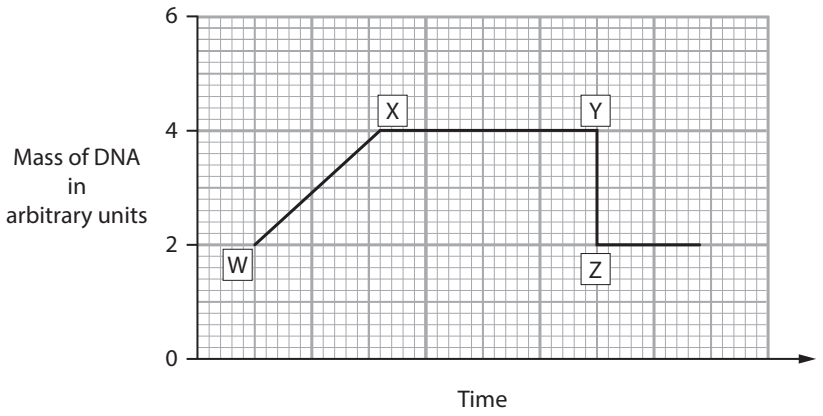
Worked-example retry · Attempt part a.iii only.

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(b) The graph shows the changes in the mass of DNA in a body cell, during mitosis.




(i) Describe the process that produces the change in the mass of DNA between points W and X.

**(4)**

(ii) Describe what is happening in the cell between points Y and Z.

**(2)**

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## 1. Question 4 · part a.iii · 3 marks · Working and answer

Working and answer

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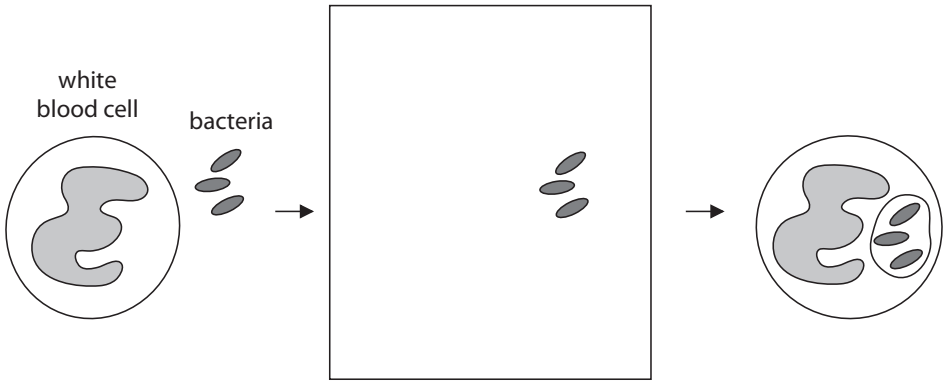
# 2. Question 2 · part c · 1 mark

June 2019 Paper 1

Further practice · Attempt part c only.

**2** (a) Bacteria can cause disease in humans.

The diagram shows how one type of white blood cell helps to defend the body against disease.




(i) Complete the diagram by drawing the shape of the white blood cell at stage 2.

**(1)**

(ii) The box lists words associated with bacteria and disease.

| | | | | | |
|---|---|---|---|---|---|
| acids | enzymes | erythrocytes | lymphocytes | phagocytes | toxins |

Use words from the box to complete the sentences.

**(2)**

White blood cells called ..... engulf bacteria.

These white blood cells contain ..... to digest bacteria.

(iii) One way that white blood cells defend the body from disease is shown in the diagram.

State another way in which white blood cells defend the body against disease.

**(1)**

.....

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# 2. Question 2 · part c · 1 mark

June 2019 Paper 1

Further practice · Attempt part c only.

(b) The table lists structures found in some cells.

Place ticks in boxes to show which structures are found in bacterial cells and which are found in human skin cells.

One row has been completed for you.

(3)

| Structures | Bacterial cell | Human skin cell |
|---|---|---|
| nucleus | | |
| DNA | | |
| cytoplasm | ✓ | ✓ |
| cell wall | | |

(c) Viruses can also cause diseases in humans.

Many viruses contain RNA as their genetic material.

Which statement describes the structure of RNA?

(1)

- ☒ **A** a double-stranded helix containing the bases ATGC
- ☒ **B** a double-stranded helix containing the bases AUGC
- ☒ **C** a single-stranded helix containing the bases ATGC
- ☒ **D** a single-stranded helix containing the bases AUGC

**(Total for Question 2 = 8 marks)**

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Working and answer

## 3. Question 2 · part b · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b only.

**2** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

**Treatment of Ebola**

Ebola is caused by an RNA virus. There is no proven treatment available for Ebola. However, whole blood collected from someone who has recovered from infection has been used as a treatment. The results have been promising in a small group of Ebola patients.

Guidance for national health authorities and blood transfusion services describes the stages needed to collect whole blood (CWB) or plasma (CP) from Ebola-recovered patients. This can be used for transfusion to patients as a treatment for early Ebola.

The guidance states how to

- identify people who have recovered from Ebola as potential blood donors
- screen donors' blood for A, B, O and Rhesus groupings
- screen donors' blood for transfusion-transmissible infections
- collect blood and care for donors
- obtain agreement of Ebola patients for the treatment
- identify patient's blood grouping
- store and transport CWB and CP to the places where transfusions are to be given
- select Ebola patients for transfusion
- transfuse the blood into the patient

Donated CWB should never be frozen and should be stored between 2 °C and 6 °C.

CWB and CP donations should be transfused to the Ebola patients using standard transfusion procedures.

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## 3. Question 2 · part b · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b only.

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(a) (i) Explain what is meant by an RNA virus (line 1).

**(2)**

(ii) The table lists five diseases. Put a tick (✓) next to the diseases that should be tested for in the donors' blood.

**(2)**

| Disease | Blood tested (✓) |
|---|---|
| anaemia | |
| cystic fibrosis | |
| gonorrhoea | |
| HIV | |
| scurvy | |

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(iii) Explain why it is necessary to identify the blood group of the blood collected from donors who have recovered from Ebola (line 9).

**(3)**

## 3. Question 2 · part b · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b only.

(iv) Suggest why donated CWB should never be frozen (line 17).

**(2)**

(v) Explain how the blood of people who have recovered from Ebola is likely to be effective in treating the disease.

**(2)**

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(b) The nitrogenous base composition of another virus was found to be adenine 13%, guanine 26%, uracil 20% and cytosine 41%.

Explain how these figures prove that this is a virus with a single strand of RNA.

**(3)**

**(Total for Question 2 = 14 marks)**

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## 3. Question 2 · part b · 3 marks · Working and answer

Working and answer

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# 4. Question 1 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

**1** (a) The diagram shows part of a molecule of DNA.




(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

**(1)**

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

**(1)**

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

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# 4. Question 1 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

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(iii) Describe how the two backbones of the DNA molecule are joined together.

**(2)**

(iv) State the name of the part of the cell where most of the DNA is found.

**(1)**

(b) DNA is a nucleic acid.

RNA is another type of nucleic acid.

(i) Give two differences between the structures of DNA and RNA.

**(2)**

1 .....

2 .....

(ii) There are different types of RNA.

Which of these are two types of RNA?

**(1)**

- ☒ **A** mRNA and pRNA
- ☒ **B** dRNA and tRNA
- ☒ **C** dRNA and pRNA
- ☒ **D** mRNA and tRNA

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## 4. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

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# 5. Question 1 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

**1** (a) The diagram shows part of a molecule of DNA.




(i) Structure X is the backbone of the DNA molecule.

Which of these form part of the backbone of DNA?

**(1)**

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

(ii) What name is given to the structures labelled Y?

**(1)**

- ☒ **A** amino acids
- ☒ **B** bases
- ☒ **C** proteins
- ☒ **D** sugars

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## 5. Question 1 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

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(iii) Describe how the two backbones of the DNA molecule are joined together.

**(2)**

(iv) State the name of the part of the cell where most of the DNA is found.

**(1)**

(b) DNA is a nucleic acid.

RNA is another type of nucleic acid.

(i) Give two differences between the structures of DNA and RNA.

**(2)**

1 .....

2 .....

(ii) There are different types of RNA.

Which of these are two types of RNA?

**(1)**

- ☒ **A** mRNA and pRNA
- ☒ **B** dRNA and tRNA
- ☒ **C** dRNA and pRNA
- ☒ **D** mRNA and tRNA

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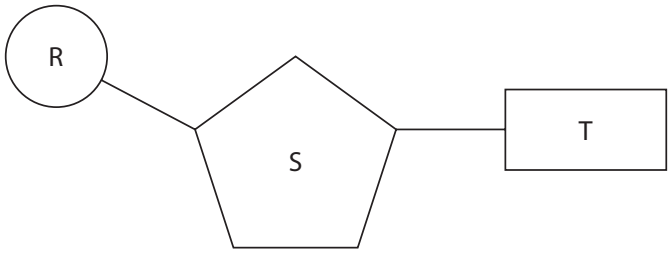
## 5. Question 1 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

(c) DNA is formed from molecules called nucleotides.

The diagram shows a single nucleotide.




Name the parts of the nucleotide labelled R, S and T.

**(3)**

R .....

S .....

T .....

**(Total for Question 1 = 11 marks)**

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Working and answer

# 6. Question 4 · part a.ii · 1 mark

November 2023 Paper 1

Further practice · Attempt part a.ii only.

**4** The mass of DNA in a body cell changes during the cell cycle.

(a) (i) Which structure in a cell contains DNA?

**(1)**

- ☒ **A** chromosome
- ☒ **B** endoplasmic reticulum
- ☒ **C** cell membrane
- ☒ **D** ribosome

(ii) Name **one** other type of nucleic acid.

**(1)**

.....

(iii) Give **three** differences between DNA and the nucleic acid that you have named in (a)(ii).

**(3)**

| Differences | DNA | Named nucleic acid |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |

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## 6. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

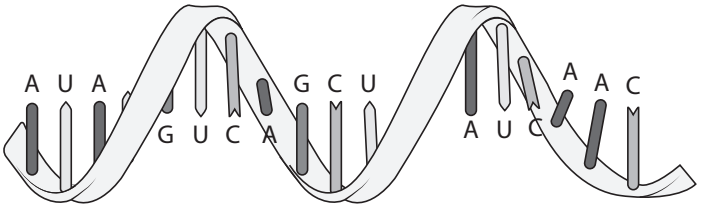
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# 7. Question 2 · part a.i · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.i only.

**2** (a) The diagram shows a strand of RNA found in a body cell.




(Source: https://upload.wikimedia.org/wikipedia/commons/3/37/Difference_DNA_RNA-EN.svg)

(i) State two places where RNA is found in a cell.

**(2)**

1 .....

2 .....

(ii) State two differences between RNA and DNA.

**(2)**

1 .....

.....

2 .....

.....

(iii) RNA and DNA are both made of repeating subunits joined in a chain.

Give the name of these repeating subunits.

**(1)**

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## 7. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

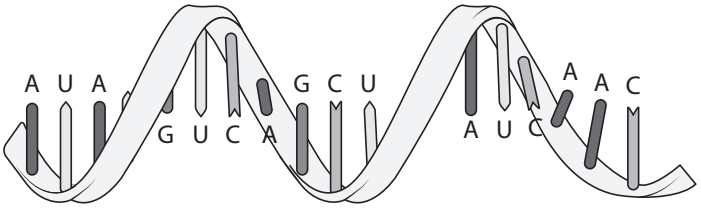
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# 8. Question 2 · part a.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.ii only.

**2** (a) The diagram shows a strand of RNA found in a body cell.




(Source: https://upload.wikimedia.org/wikipedia/commons/3/37/Difference_DNA_RNA-EN.svg)

(i) State two places where RNA is found in a cell.

**(2)**

1 .....

2 .....

(ii) State two differences between RNA and DNA.

**(2)**

1 .....

.....

2 .....

.....

(iii) RNA and DNA are both made of repeating subunits joined in a chain.

Give the name of these repeating subunits.

**(1)**

.....

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## 8. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

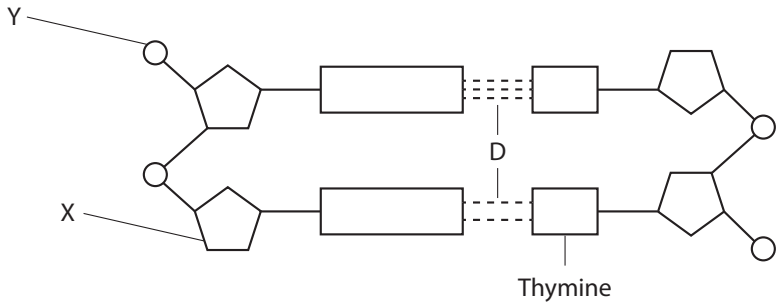
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# 9. Question 2 · part a.ii · 1 mark

June 2025 Paper 1

Further practice · Attempt part a.ii only.

**2** The diagram shows part of a molecule of nucleic acid.




(a) (i) Name the parts labelled X and Y.

**(2)**

X .....

Y .....

(ii) Name the type of nucleic acid shown in the diagram.

**(1)**

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(iii) Use information from the diagram to justify your answer to part (ii).

**(3)**

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(iv) State where this nucleic acid is found in a human cell.

**(1)**

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## 9. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

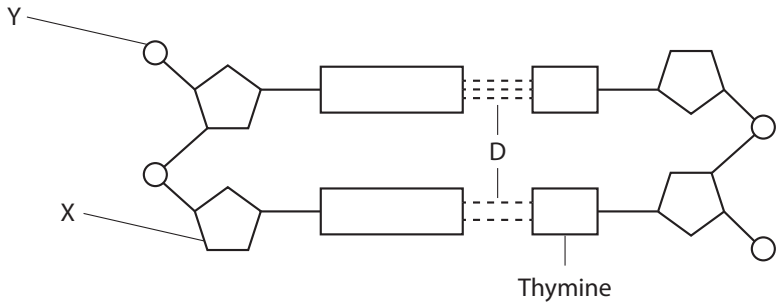
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# 10. Question 2 · part a.iii · 3 marks

June 2025 Paper 1

Further practice · Attempt part a.iii only.

**2** The diagram shows part of a molecule of nucleic acid.




(a) (i) Name the parts labelled X and Y.

**(2)**

X .....

Y .....

(ii) Name the type of nucleic acid shown in the diagram.

**(1)**

.....

(iii) Use information from the diagram to justify your answer to part (ii).

**(3)**

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(iv) State where this nucleic acid is found in a human cell.

**(1)**

.....

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## 10. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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